Industrial Internship Report on

”Student Dropout Prediction”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was **Student Dropout Prediction** This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

Summary:

During the 6-week internship at Uniconverge Technologies (USC/UCT), the program focused on building a strong foundation in Data Science and Machine Learning through a blend of theoretical learning and practical implementation.

- Week 1–2: Introduction to Python for Data Science, data handling using Pandas and NumPy, and basic data preprocessing techniques.
- Week 3–4: Core Machine Learning algorithms such as Linear Regression, Logistic Regression, K-Means Clustering, and Decision Trees were studied and implemented.
- Week 5: Model evaluation techniques, feature engineering, and exploratory data analysis (EDA) were performed on real-world datasets.
- Week 6: Final project development, model optimization, and presentation of results.

Overall, the internship emphasized hands-on learning, dataset-driven problem solving, and practical exposure to industry workflows.

Planning:

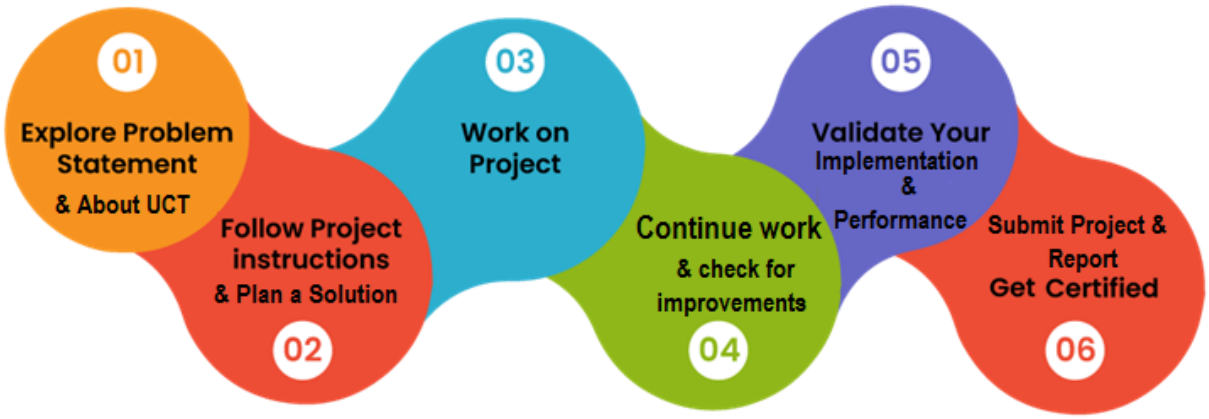
The internship program was systematically designed to ensure progressive learning:

- Step-by-step curriculum: Starting from basics to advanced ML concepts
- Weekly milestones: Each week focused on specific learning objectives
- Practical implementation: Every concept was followed by coding exercises and case studies
- Project-based learning: Final weeks dedicated to applying all learned skills
- Mentorship & evaluation: Regular guidance, feedback, and performance assessment

The structured approach ensured gradual skill development, reinforcement through practice, and final application via project work.

Conclusion

The 6-week internship at Uniconverge Technologies provided a comprehensive introduction to Data Science and Machine Learning. It not only enhanced technical proficiency but also improved problem-solving ability and industry readiness, making it a valuable step toward a successful career in the field.



Learnings and Overall Experience

The internship provided valuable hands-on exposure to Data Science and Machine Learning. I gained practical knowledge in data preprocessing, EDA, and implementation of ML algorithms using Python libraries. It enhanced my analytical thinking, problem-solving ability, and understanding of real-world data-driven workflows.

Acknowledgement

I sincerely thank the team at **Uniconverge Technologies (USC/UCT)** for this opportunity. I am grateful to my mentors, faculty members, friends, and family for their continuous guidance and support throughout the internship.

.**Message to Juniors and Peers**

Focus on building strong fundamentals, work on practical projects, and make the most of internship opportunities. Consistency and continuous learning are key to success in this field.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

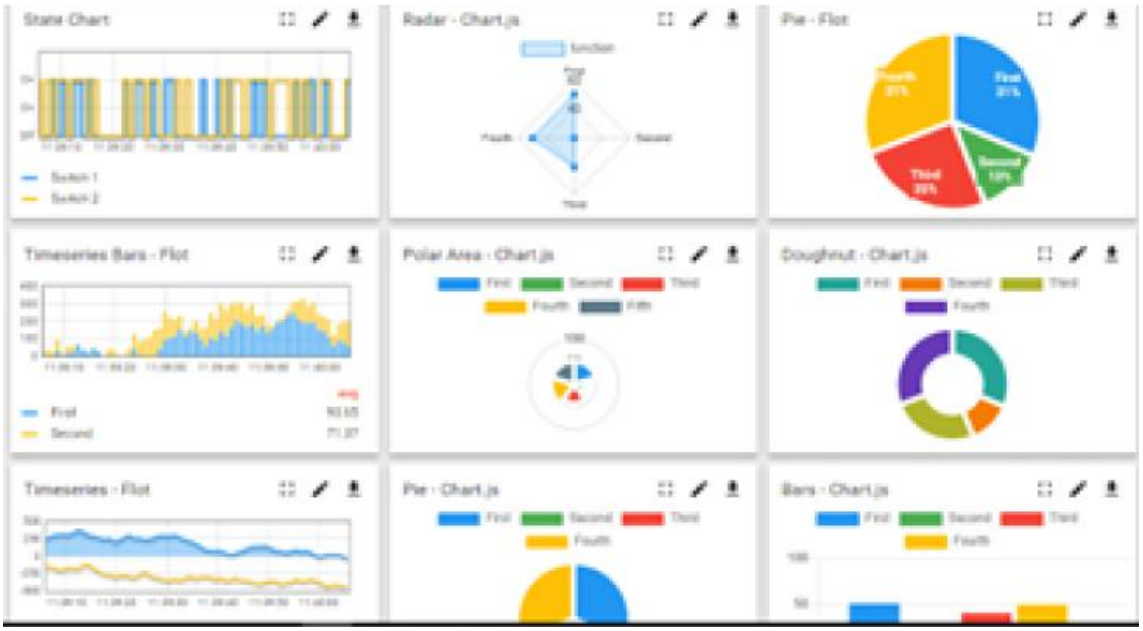
i. UCT IoT Platform (uct Insight)

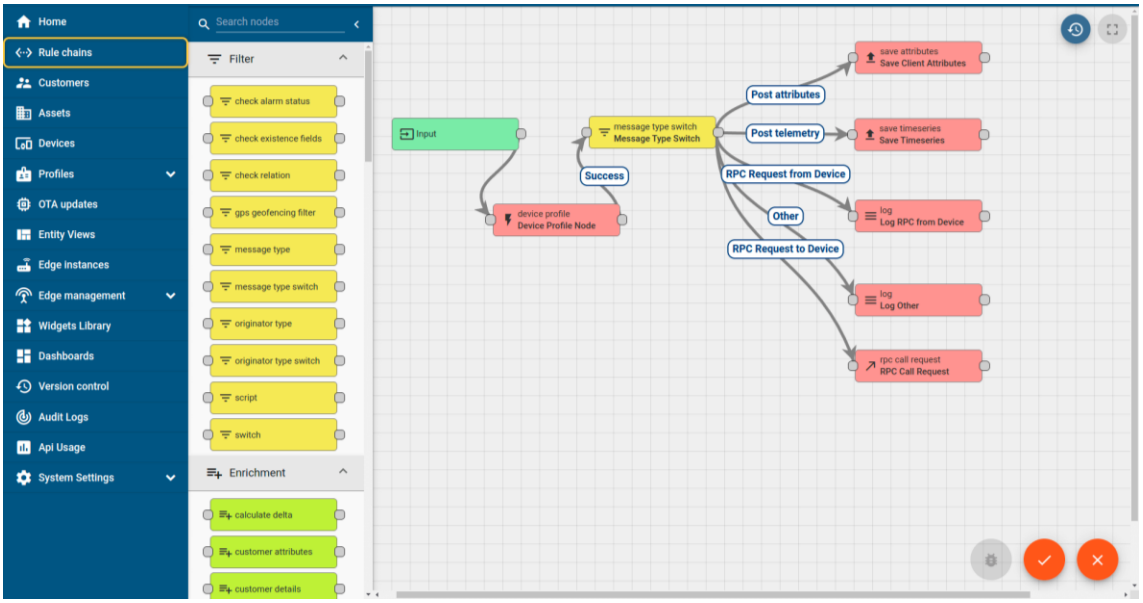
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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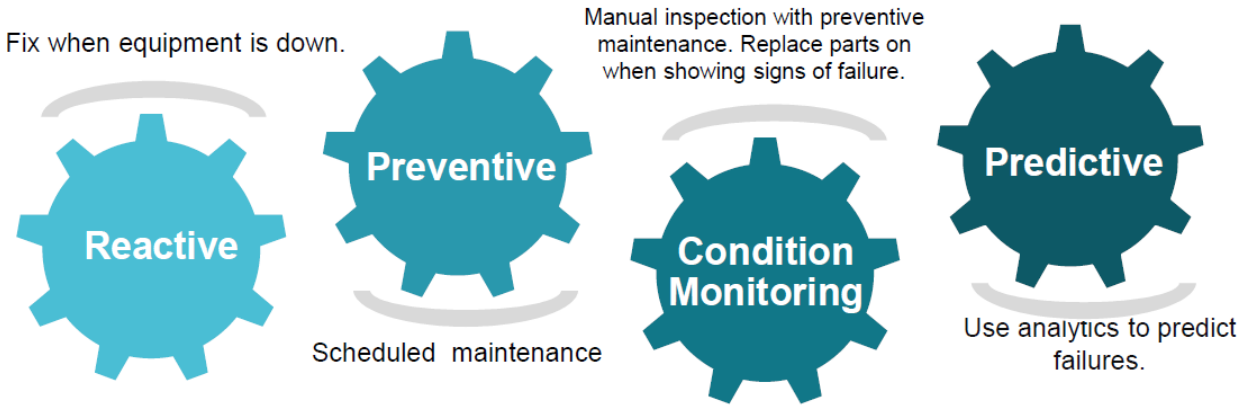
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

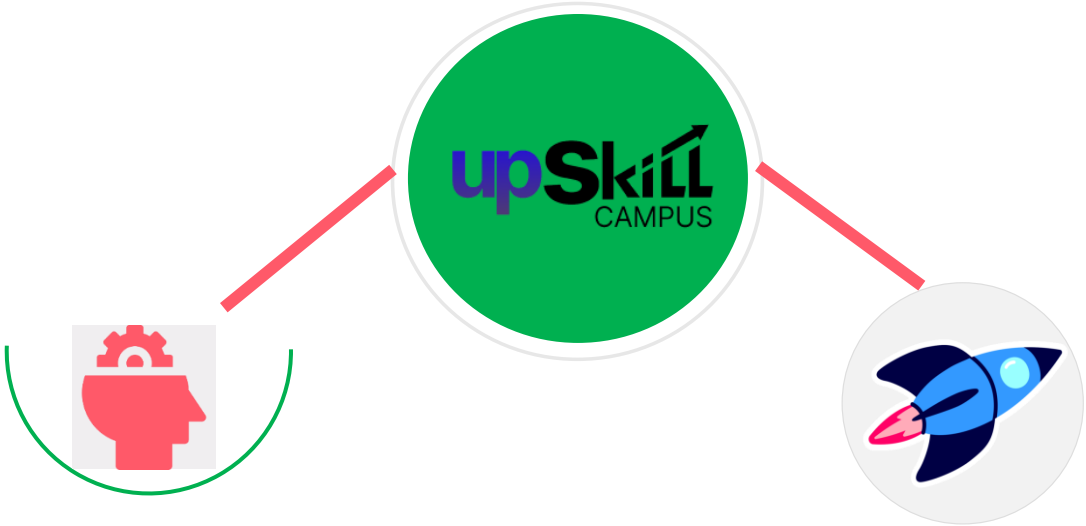
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

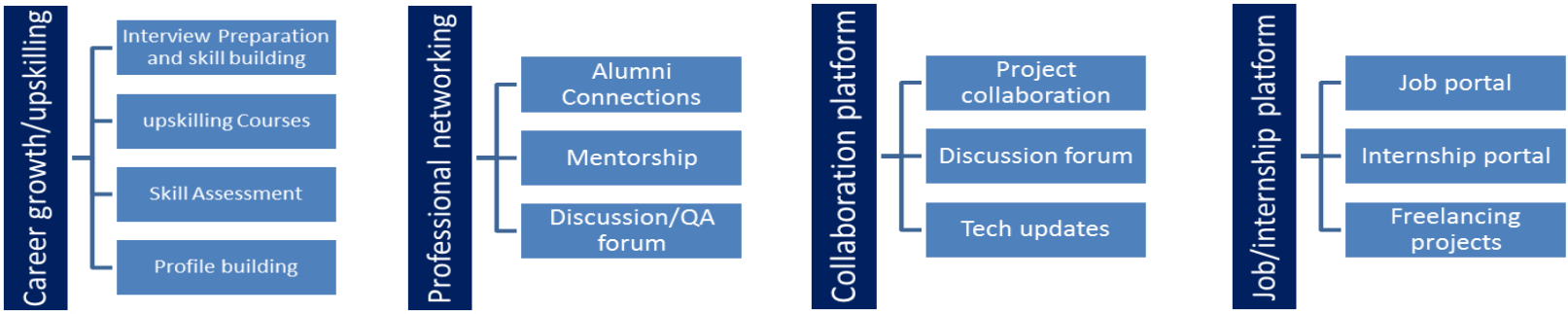
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Documentation of **Python**, **Pandas**, **NumPy**, and **Scikit-learn** libraries

[2] Official resources and training materials provided by **Uniconverge Technologies (USC/UCT)**

[3] Online learning platforms and tutorials related to Data Science and Machine Learning

2.6 Glossary

Terms	Acronym
Data Science	D**S**
Machine Learning	M**L**
Artificial Intelligence	A**I**
Exploratory Data Analysis	E**DA**
Feature Engineering	FE

3 Problem Statement

In the assigned problem statement

Student dropout is a critical challenge faced by educational institutions, leading to academic loss and reduced institutional performance. Identifying students at risk of dropping out at an early stage is difficult due to the involvement of multiple academic, social, and behavioral factors.

The problem is to develop a Machine Learning-based predictive system that analyzes student-related data and accurately classifies whether a student is at high risk or low risk of dropout. The system should also identify key influencing factors and provide interpretable insights to support timely intervention by educators

4 Existing and Proposed solution

Existing Solutions and Limitations

Existing approaches to student dropout prediction primarily use traditional statistical methods and basic machine learning models such as Logistic Regression and Decision Trees. Some systems also rely on manual analysis by educators.

Limitations:

- Limited accuracy due to use of simple models and small datasets
- Inability to capture complex relationships between multiple factors
- Lack of real-time prediction and automation
- Poor interpretability in some advanced models or over-simplification in basic ones
- Limited focus on actionable insights for early intervention

Proposed Solution

The proposed system is a **Machine Learning-based Student Dropout Prediction System** that uses structured student data to predict dropout risk.

- Implementation of **Logistic Regression** for probability-based prediction
- Use of **Decision Tree** for interpretable classification and rule extraction
- Data preprocessing and feature engineering to improve model performance
- Evaluation using appropriate performance metrics (accuracy, precision, recall)

4.1 Code submission (Github link)

https://github.com/SoumyadeepChattopadhyay2004/upSkillCampus/blob/main/Student_Dropout_Prediction_System.ipynb

4.2 Report submission (Github link) :

https://github.com/SoumyadeepChattopadhyay2004/upSkillCampus/blob/main/Student_Dropout_Prediction_System.pdf

5 Proposed Design/ Model

The proposed system follows a structured **end-to-end Machine Learning pipeline** consisting of the following stages:

1. Data Collection

The dataset is obtained from the **UCI Student Performance Dataset**, containing academic, social, and behavioral attributes of students.

2. Data Preprocessing

- Handling missing values and inconsistencies
- Encoding categorical variables
- Normalization/standardization (if required)
- Creation of target variable (**Dropout Risk**) based on final grades

3. Exploratory Data Analysis (EDA)

- Identifying patterns and correlations between features
- Visualizing important factors such as study time, absences, and grades
- Detecting outliers and data distribution

4. Feature Engineering

- Selecting relevant features influencing dropout
- Transforming variables to improve model performance

5. Model Building

- **Logistic Regression:** Used for probability-based prediction of dropout risk
- **Decision Tree:** Used for classification and generating interpretable decision rules

6. Model Evaluation

- Performance measured using **accuracy, precision, recall, and confusion matrix**

- Comparison of both models to select the best-performing approach

7. Prediction and Output

- The final model predicts whether a student is at **Low Risk (0)** or **High Risk (1)**
- Provides insights into key factors contributing to dropout

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

Decision Tree Accuracy:

0.759493670886076

Classification Report

	precision	recall	f1-score	support
0	0.74	0.98	0.84	52
1	0.90	0.33	0.49	27
accuracy			0.76	79
macro avg	0.82	0.66	0.66	79
weighted avg	0.79	0.76	0.72	79

Confusion Matrix

[[51 1]

[18 9]]

6.1 Test Plan/ Test Cases

Test Case	Input Condition	Expected Output
TC1	Student with high grades and low absences	Low Risk (0)
TC2	Student with low grades (<10)	High Risk (1)
TC3	Missing/incorrect data	Handled during preprocessing

Test Case	Input Condition	Expected Output
TC4	Mixed attributes (moderate grades, high absences)	Model prediction based on learned pattern
TC5	New unseen data	Accurate classification based on trained model

6.2 Test Procedure

- Split dataset into **training and testing sets**
- Train models (Logistic Regression, Decision Tree) on training data
- Apply model on test data
- Evaluate using metrics like **accuracy, precision, recall, confusion matrix**
- Compare results of both models
- Validate predictions using sample inputs

6.3 Performance Outcome

- The model successfully classifies students into **High Risk and Low Risk** categories
- Logistic Regression provides **probability-based prediction**
- Decision Tree offers **interpretable decision rules**
- Achieved satisfactory performance in terms of **accuracy and reliability**
- Model is effective for **early identification of at-risk students**

7 My learnings

This project enhanced my understanding of the complete Machine Learning pipeline, including data preprocessing, model building, and evaluation. It improved my analytical thinking and problem-solving skills. The experience will help me in building real-world ML solutions and strengthen my career in Data Science.

8 Future work scope

- Implementation of advanced models like **Random Forest, XGBoost, or Neural Networks**
- Integration with **real-time student data systems**
- Development of a **web-based dashboard** for visualization
- Inclusion of more diverse datasets for better generalization
- Deployment of the model for **practical institutional use**